

Storing Memories

After being encoded by the hippocampus, memories are consolidated and transferred to the cortex for long-term storage. These memories are formed by strengthening connections, a process called long-term potentiation (see pp.136–137).

Storage in the cortex

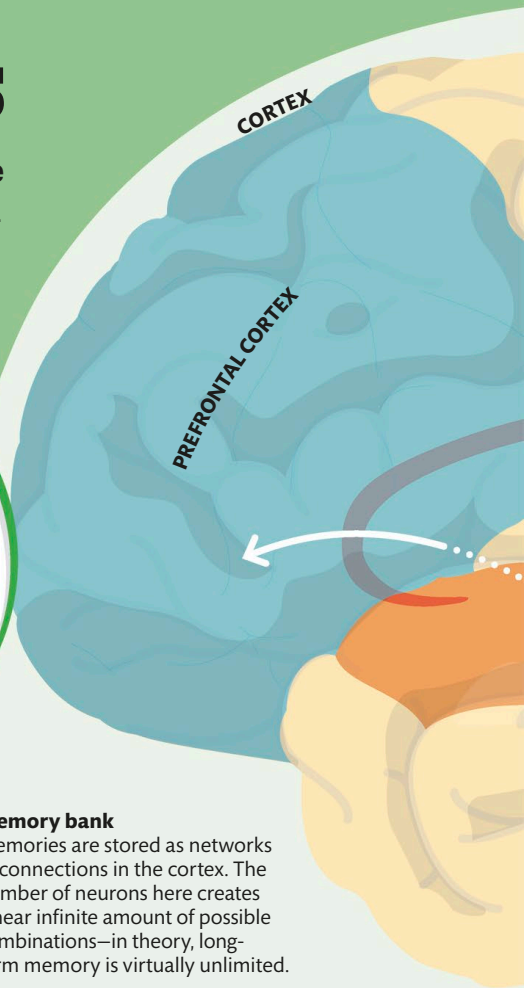
To transfer memories for long-term storage, the hippocampus repeatedly activates a network of connections in the cortex. Each activation strengthens connections until they are secure enough to store the memory. It was thought that memories formed first in the hippocampus, with the cortical memory trace forming later, but recent research in mice suggests that they may form simultaneously, although the cortical memory is initially unstable. Repeated reactivation of the network somehow “matures” the cortical memory, meaning we can use it.

WHY DO I FORGET WHERE I LEFT MY KEYS?

Often, things we “forget” actually weren’t stored as memories in the first place, because we weren’t paying attention when we did them.

Memory bank

Memories are stored as networks of connections in the cortex. The number of neurons here creates a near infinite amount of possible combinations—in theory, long-term memory is virtually unlimited.



Consolidation

This storage process, known as consolidation, happens mainly while we sleep. During this time, your brain is not processing information from the outside world so it can carry out these housekeeping tasks. Memories are sorted, prioritized, and the gist extracted. They are also linked with older memories, already in storage. This makes it easier to retrieve important memories in the future. Studies have shown it is better to take a nap after learning something new than it is to keep studying!

LEARNING

1

Study

When you learn something new, your brain takes in that information and forms new connections, or strengthens synapses that already exist.



CONSOLIDATION

2

Sleep

While you sleep, new information is consolidated. The memory becomes less reliant on the hippocampus and less likely to be affected by interference from other inputs or brain injury.

